

HTML Forms & Semantic HTML

Today we dive into two essential pillars of web development: **HTML Forms** — the gateway to collecting user data — and **Semantic HTML**, which gives your web pages meaningful structure. By the end of today, you'll be able to build a real student registration form and a class timetable from scratch.

HTML Forms

Input types, labels, buttons, textarea, select & option

Tables

Organize data into structured rows and columns

Semantic HTML

Header, footer, nav, section, article, aside

Practical Tasks

Student Registration Form + Timetable using Table

"Forms collect user information from websites — they are the bridge between your user and your server."

Introduction to HTML Forms

An HTML **form** is a section of a webpage that allows users to enter and submit data. Every time you log in, sign up, or fill out a contact page — you're using a form. Forms are one of the most critical building blocks of interactive websites.

Form Syntax

All form elements are wrapped inside a `<form>` tag.

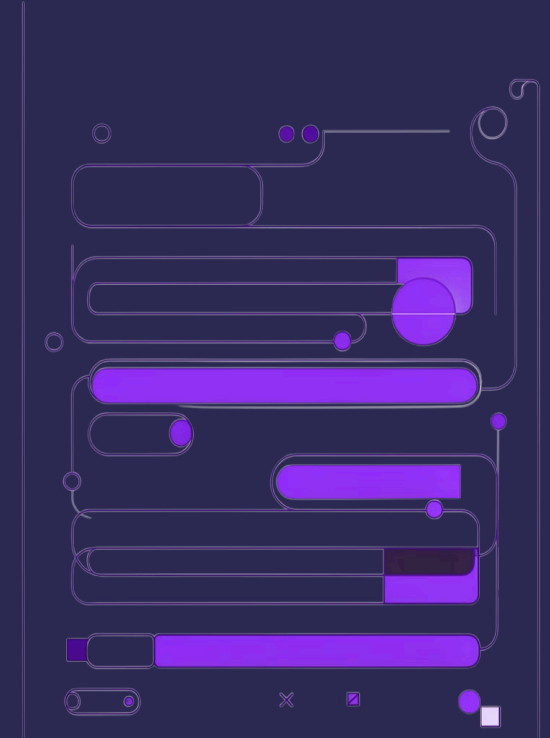
```
<form>
  <!-- inputs go here -->
</form>
```

Why Forms Matter

- **User Registration** — Create accounts on websites
- **Login Systems** — Authenticate returning users
- **Online Applications** — Submit forms to institutions
- **Data Collection** — Gather feedback and survey responses

Common Real-World Examples

- Google Sign-In form
- Facebook registration
- College admission portal
- Contact us page



Input Types in HTML

The `<input>` tag is the most versatile element in HTML forms. By changing the `type` attribute, you control what kind of data the user can enter. Each type comes with built-in browser validation and behavior — no extra code needed!

1

text

```
<input type="text"
placeholder="Enter Name">
```

Accepts any plain text input

2

email

```
<input type="email"
placeholder="you@example.com">
```

Validates proper email format automatically

3

password

```
<input type="password">
```

Hides characters as the user types

4

radio & checkbox

```
<input type="radio">
<input type="checkbox">
```

Single or multiple choice selections

 Other useful types include `type="number"` for numeric input and `type="date"` to open a calendar date picker in the browser.

Labels & Buttons

Labels — Describe Your Inputs

The `<label>` tag tells the user what to type in each field. Good labels make your form accessible and easier to understand. Always pair a label with its corresponding input field.

```
<label>Full Name</label>  
<input type="text">
```

- ✔ Linking a label to an input using the `for` attribute lets users click the label to focus the field — a big usability win!

Buttons — Trigger Actions

Buttons let users submit, reset, or interact with a form. There are three main types:

- **Submit** — sends form data to the server
- **Reset** — clears all inputs back to default
- **Button** — custom action (used with JavaScript)

```
<button type="submit">  
  Register  
</button>
```

```
<input type="reset">
```

Textarea, Select & Option

Some inputs need more than a single line of text or a simple value. The `<textarea>` and `<select>` elements handle those cases beautifully — and you'll use them in almost every real-world form.



Textarea

Used for collecting longer text such as addresses, messages, or feedback. The `rows` and `cols` attributes control its size.

```
<textarea  
  rows="4"  
  cols="30">  
</textarea>
```



Select & Option

Creates a dropdown menu. The `<select>` tag is the container; each `<option>` tag is one choice in the list.

```
<select>  
  <option>CSE</option>  
  <option>ECE</option>  
  <option>EEE</option>  
</select>
```



Real-World Uses

These elements appear constantly in everyday web applications across all industries.

- Country & city selection
- Course or department selection
- Shipping address entry
- Feedback & review boxes

Complete Form Example

Student Registration Form

This example brings together everything we've learned — inputs, labels, a dropdown, a textarea, and a submit button. Study how each element is used together to create a complete, functional form.

```
<form>
  <label>Name:</label>
  <input type="text"><br><br>

  <label>Email:</label>
  <input type="email"><br><br>

  <label>Password:</label>
  <input type="password"><br><br>

  <label>Department:</label>
  <select>
    <option>CSE</option>
    <option>ECE</option>
  </select><br><br>

  <label>Address:</label>
  <textarea></textarea><br><br>

  <button>Register</button>
</form>
```

Concepts Used

Input Fields

Text, email, and password types

Labels

Describe each field clearly

Dropdown

Select + Option for department

Textarea

Multi-line address field

Button

Submits the completed form

Introduction to HTML Tables

Tables in HTML are used to display structured data in a grid of **rows and columns**. Think of them like a spreadsheet embedded in your webpage — perfect for timetables, comparison charts, and data summaries.

Key Table Tags

→ **<table>**

The outer container for the entire table

→ **<tr>** – Table Row

Creates a horizontal row inside the table

→ **<th>** – Table Heading

Bold, centered column headers

→ **<td>** – Table Data

Regular cells containing your content

Example Code

```
<table border="1">
  <tr>
    <th>Day</th>
    <th>Subject</th>
    <th>Time</th>
  </tr>
  <tr>
    <td>Monday</td>
    <td>HTML</td>
    <td>9:00 AM</td>
  </tr>
  <tr>
    <td>Tuesday</td>
    <td>CSS</td>
    <td>10:00 AM</td>
  </tr>
</table>
```

❗ The `border="1"` attribute adds a visible border around all cells — great for learning and debugging your table structure.

What is Semantic HTML?

Semantic HTML means using tags that **describe the meaning** of their content — not just how it looks. Instead of using generic `<div>` tags for everything, semantic tags tell both the browser and developers exactly what role each section plays on the page.

`<header>`

Top section — site logo, title, or intro banner

`<nav>`

Navigation links — menus and page links

`<section>`

A standalone thematic block of content

`<article>`

Self-contained content like a blog post

`<aside>`

Side content — related links or callouts

`<footer>`

Bottom section — copyright, contact info

4X

Better SEO

Search engines rank semantic pages higher

100%

Accessibility

Screen readers understand your page structure



Readability

Easier for any developer to read your code

Semantic Tags in Practice

Here's how semantic HTML tags work together to build a well-structured webpage. Notice how each tag's name tells you exactly what it contains — making the code much easier to read and maintain than a page full of `<div>` tags.

Page Structure Code

```
<header>
  <h1>My Website</h1>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Contact</a>
</nav>

<section>
  <article>
    <h2>HTML Basics</h2>
    <p>Learn the fundamentals...</p>
  </article>
</section>

<aside>
  <p>Related: CSS Tutorial</p>
</aside>

<footer>
  <p>Copyright 2026</p>
</footer>
```

Visual Page Layout

Think of a webpage as a newspaper — each section has a clear, named role:



`<header>`

The masthead — your brand at the very top



`<nav>`

The table of contents — your site's menu



`<section> + <article>`

The main story — your core page content



`<footer>`

The back page — copyright and contact info

Practical Tasks & Homework

Now it's time to put your knowledge to work! Complete both in-class tasks today, then finish the homework assignment before next class. Practice is the fastest way to build confidence with HTML forms.

TASK 1 — IN CLASS

Student Registration Form

- Name (*text input*)
- Email (*email input*)
- Password (*password input*)
- Gender (*radio buttons*)
- Department (*select dropdown*)
- Address (*textarea*)
- Submit Button

TASK 2 — IN CLASS

Class Timetable using Table

- Days of the week (*rows*)
- Subject names (*columns*)
- Timings for each session
- Use `<th>` for headers
- Add `border="1"` for clarity

HOMEWORK

College Admission Form

- Student Name & Father's Name
- Mobile Number & Email
- Course selection (*dropdown*)
- Address (*textarea*)
- Submit Button

"Practice forms daily to become confident in HTML — the more you build, the more natural it feels."